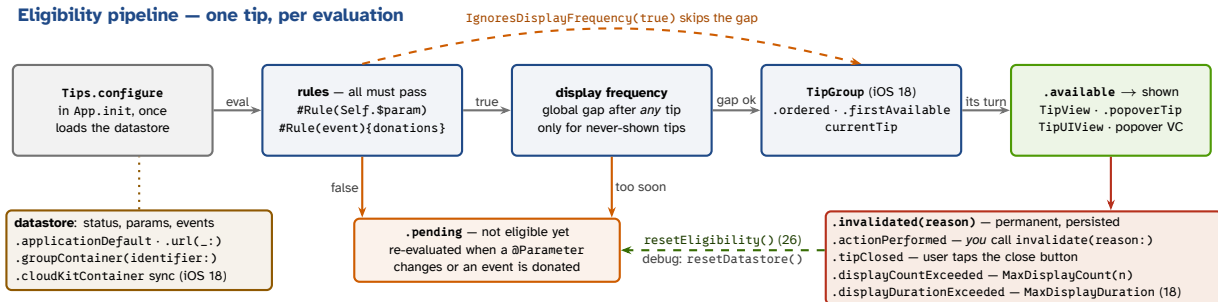


In one line: A **tip** is a small, transient view that teaches *one* non-obvious feature. You declare **content** + **eligibility** on a Tip; TipKit (iOS 17) keeps state in a persistent **datastore**, evaluates **rules** (@Parameter values, counted Tips.Event donations), spaces tips by a global **display frequency**, and retires a tip for good once **invalidated** — user acted, closed it, or it hits its cap.

Eligibility pipeline — one tip, per evaluation



How it works

- **Tip** (Identifiable, Sendable): title, message?, image?, actions, rules, options — the last three are result builders.
- **Tips.configure(_)** (throws) *must* run before any tip displays. Frequency .immediate (default) · .hourly · .daily · .weekly · .monthly.
- **Parameter rule** = app state. @Parameter values are **persisted** across launches (.transient resets); assigning one re-evaluates the rules that read it.
- **Event rule** = repeated behaviour: await event.donate() (sendDonation() from sync code); predicate over \$0.donations: .donatedWithin(.days(5)), .count, largestSubset(groupedBy:). Event<Info: Codable> donations carry data to filter on. Queries see the latest **1000** donations; iOS 26: DonationLimit(maximumCount:maximumAge:).
- **Display frequency** = minimum gap after a tip shows before *another, never-shown* tip may appear. A tip already shown still reappears while its rules pass.
- **Options:** MaxDisplayCount(3) (no default cap) · MaxDisplayDuration(300) (iOS 18, cumulative, ≥ 60 s) · IgnoresDisplayFrequency(true).
- **Observe:** status · shouldDisplay · statusUpdates · shouldDisplayUpdates. SwiftUI views observe for you; UIKit code must.

Example — the tip

```
import TipKit
struct ShareTip: Tip {
    @Parameter static var isLoggedIn: Bool = false // persisted
    static let openedDoc = Tips.Event(id: "openedDoc") // counted
    var title: Text { Text("Share with one tap") }
    var message: Text? { Text("Send this document to a teammate.") }
    var image: Image? { Image(systemName: "square.and.arrow.up.fill") }
    var actions: [Action] { Action(id: "how", title: "Show me") }
    var rules: [Rule] { // AND: each narrows
        #Rule(Self.$isLoggedIn) { $0 == true }
        #Rule(Self.openedDoc) { $0.donations.donatedWithin(.week).count >= 3 }
    }
    var options: [any TipOption] { MaxDisplayCount(3) }
}
```

Example — configure, feed, present, retire

```
@main struct DocsApp: App {
    init() {
        #if DEBUG
            try? Tips.resetDatastore() // BEFORE configure
        #endif
        try? Tips.configure([.displayFrequency(.daily)])
    }
    var body: some Scene { WindowGroup { EditorView() } }
    // login: ShareTip.isLoggedIn = true -> rules re-evaluate
    // editor: .task { await ShareTip.openedDoc.donate() }
    Button("Share", systemImage: "square.and.arrow.up") {
        tip.invalidate(reason: .actionPerformed) // they found it
        share()
    }
    .popoverTip(tip) { action in if action.id == "how" { showHelp() } }
```

Presentation

API	Behaviour
TipView(tip)	inline, takes layout space ; keeps context visible anchored to a control, keeps content flow
.popoverTip(tip)	UIKit inline — you add / remove it
TipUIView	UIKit popover on a sourceItem; you present / dismiss it
TipUIPopoverViewController	a tip as a list cell · AppKit: TipNSView, TipNSPopover
TipUICollectionViewController	custom look; built-in MiniTipViewStyle

Testing

All run **before** configure; precedence
 showTipsForTesting([A.self]) > hideTipsForTesting >
 showAllTipsForTesting() > hideAllTipsForTesting().
 resetDatastore() = first-launch state. Launch args:
 -com.apple.TipKit.ResetDatastore 1,
 -com.apple.TipKit.ShowTips A,B.

When a tip is the right tool (HIG)

Simple features (≤ **3 actions**), **1–2 sentences**, actionable. Not for tours, promotion or critical info — a tip can be missed. Rule it to the people who benefit; pace several (e.g. once per 24 h). Filled SF Symbol; don't repeat the image the tip points at.

Traps

- ! No Tips.configure → no tip and **no error**.
- ! Invalidation is **persisted**: closed once in a test run, gone next run — reset the store.
- ! Forgetting invalidate(reason: .actionPerformed) nags people who already use the feature.
- ! Frequency never throttles an already-shown tip: MaxDisplayCount.
- ! UIKit: observe in a Task, cancel it in viewWillDisappear.
- ! CloudKit sync (iOS 18): a **separate** container; iCloud + Background Modes (remote notifications).
- ! TipGroup is iOS 18 (announced as TipSequence); .ordered waits for each earlier tip's invalidation.

Remember

“Configure once · rule who · pace when · cap how often · invalidate when learned.”